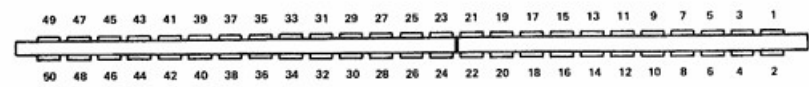
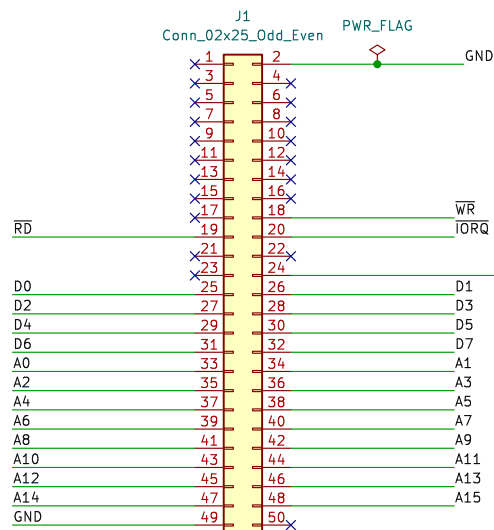


Female sockets on connector have pins reversed by orientation on the board so keep the same logical connections i.e. Pin 2 is Pin 49 and pin 3 is pin 48

VIEWED FROM REAR

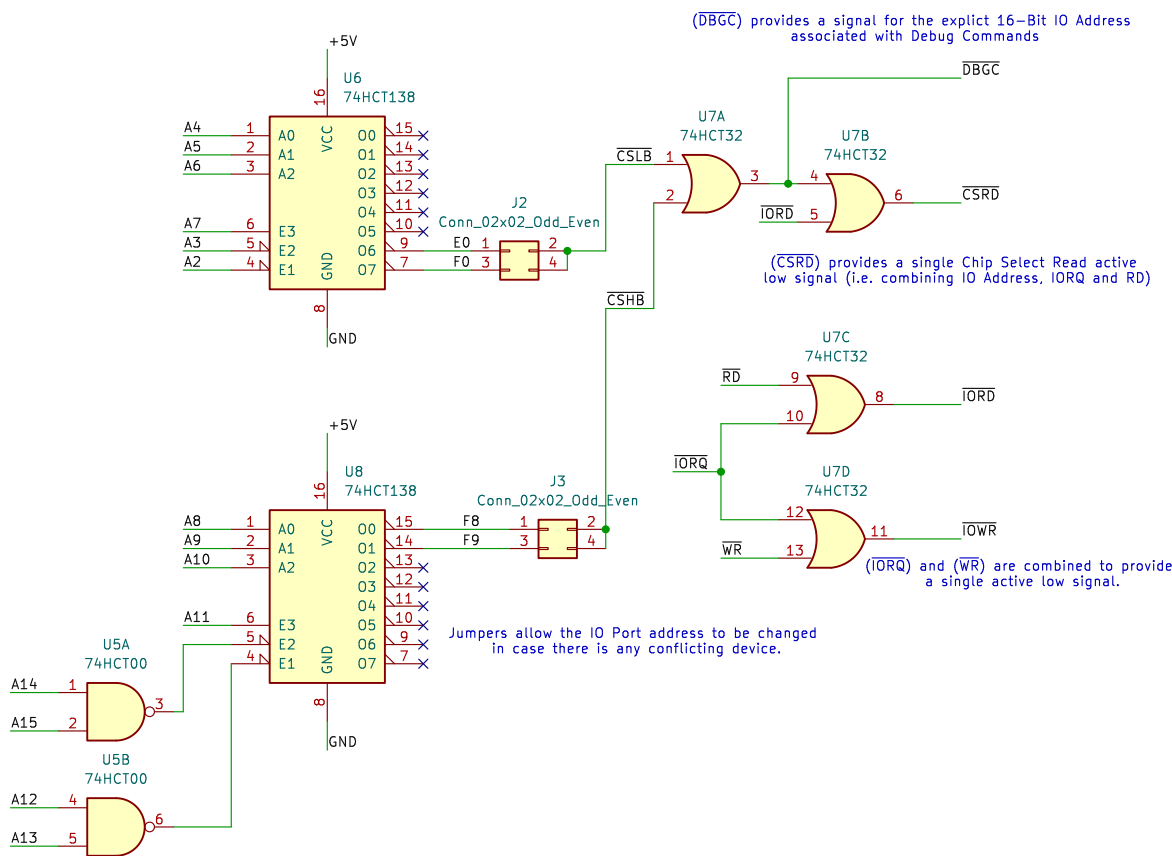


PIN 1	SOUND	PIN 18	A0	PIN 35	INT
PIN 2	GND	PIN 19	D7	PIN 36	NMI
PIN 3	A15	PIN 20	D6	PIN 37	BUSR2
PIN 4	A14	PIN 21	D5	PIN 38	BUSAK
PIN 5	A13	PIN 22	D4	PIN 39	READY
PIN 6	A12	PIN 23	D3	PIN 40	BUS RESET
PIN 7	A11	PIN 24	D2	PIN 41	RESET
PIN 8	A10	PIN 25	D1	PIN 42	ROMEN
PIN 9	A9	PIN 26	D0	PIN 43	ROMDIS
PIN 10	A8	PIN 27	+5V	PIN 44	RAMRD
PIN 11	A7	PIN 28	MREQ	PIN 45	RAMDIS
PIN 12	A6	PIN 29	M1	PIN 46	CURSOR
PIN 13	A5	PIN 30	RFSH	PIN 47	L PEN
PIN 14	A4	PIN 31	IOB0	PIN 48	EXP
PIN 15	A3	PIN 32	RD	PIN 49	GND
PIN 16	A2	PIN 33	WR	PIN 50	φ
PIN 17	A1	PIN 34	HALT		



138 Decoders provide a low and high address byte selection signal (i.e. F8E0). This is used to provide single active low signal for a port read when combined with IORQ+RD.

ADDRESS LINES															HEX ADDRESS	
INPUITS																
OUTPUTS																
ENABLE		SELECT			ADDRESS											
G1	/G2	/G2A	C	B	A	Y0	Y1	Y2	Y3	Y4	Y5	Y6	Y7			
X	1	X	X	X	X	1	1	1	1	1	1	1	1	1		
X	X	X	X	X	X	1	1	1	1	1	1	1	1	1		
1	0	0	0	0	0	1	1	1	1	1	1	1	1	1	80	
1	0	0	0	0	0	1	1	0	1	1	1	1	1	1	90	
1	0	0	0	0	1	0	1	1	0	1	1	1	1	1	A0	
1	0	0	0	0	1	1	1	0	1	0	1	1	1	1	B0	
1	0	0	0	1	0	0	1	1	1	1	0	1	1	1	CD	
1	0	0	0	1	0	1	1	1	1	1	0	1	1	1	DD	
1	0	0	0	1	1	1	1	1	1	1	1	0	1	1	ED	
1	0	0	0	1	1	1	1	1	1	1	1	1	0	1	FD	
PINS																
6	15	14	13	12	11	15	14	13	12	11	10	9	8	7		

[illegible]

The left diagram shows a parallel circuit with three capacitors, labeled C1, C2, and C3, each with a value of 100nF. They are connected between a 3V3 supply and a common ground (GND).

The right diagram shows a parallel circuit with four capacitors, labeled C4, C5, C6, and C7, each with a value of 100nF. They are connected between a +5V supply and a common ground (GND).

[CPC Wiki Amstrad CPC Default IO Ports](#)
[CPC Wiki Amstrad CPC IO Port Summary](#)
[CPC Wiki Amstrad CPC Expansion Port](#)

Bus Transceiver provide voltage level shift from 5V to 3V3 and Bus isolation

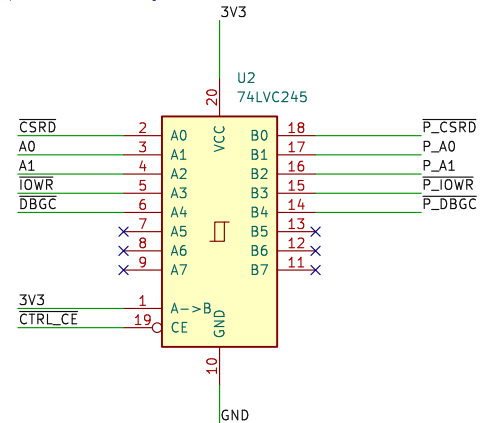
This transceiver buffers the control signals associated with IO Reads and Writes from the CPC.

Chip Select Read (**CSRD**) is active low when the CPC is reading the Pico IO port i.e. F9E0. The Pico will use this as a trigger to place the next data byte on the data lines and change direction of the Data Transceiver.

A0 and A1 have also been provided as an optional address selection and may allocate these to return the CRTC, GA or PIO data but not decided yet.

IOWR active is low when the CPC is executing any IO Write, which will include the CRTC, GA or PIO. The Pico will use this as a trigger to capture the data.

(**DBGCE**) will be pulled low when a Debug Command (DBGCE) is written to the Pico IO Port i.e. F9E0. This then ensures that only an explicit 16-bit address OUT instruction will be processed for Debug Commands.



Note Direction Fixed as A to B

This transceiver buffers the high byte of the IO Write from the CPC. Only the High Byte of the Address is used on the Amstrad CPC to select the CRTC, GA and PIO.

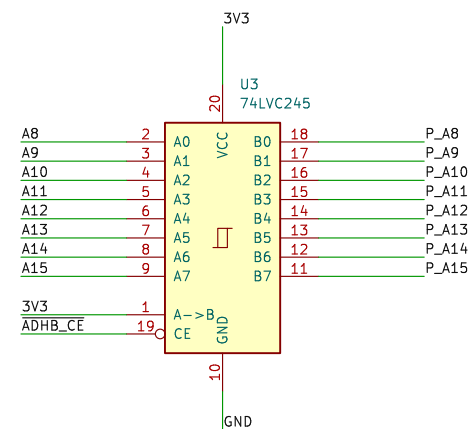
Note on the Amstrad CPC selection of the CRTC, GA and PIO is made with a single active low address line (i.e. B14 low selects the CRTC). Address High Byte Chip Enable ($\overline{\text{ADHB_CE}}$) will be pulled low by the Pico when powered.

Components use partial address decoding. It is therefore possible to send the same value to different components **simultaneously** by carefully using custom I/O ports.

[illegible]

Legend

- * "x" means this bit is ignored,
- * "0" means the bit must be set to "0" for the hardware device to respond,
- * "1" means the bit must be set to "1" for the hardware device to respond,
- * "1" and "0" mean a bit used to define a register
- * *** Bit14 of the PAL port must be at 1 on CPC's equipped with CRTCS 0/1/2. It can be at 0 or 1 on CRTCS 3/4. For compatibility reasons, it is strongly advised to always set bit14 to 1 to select PAL.

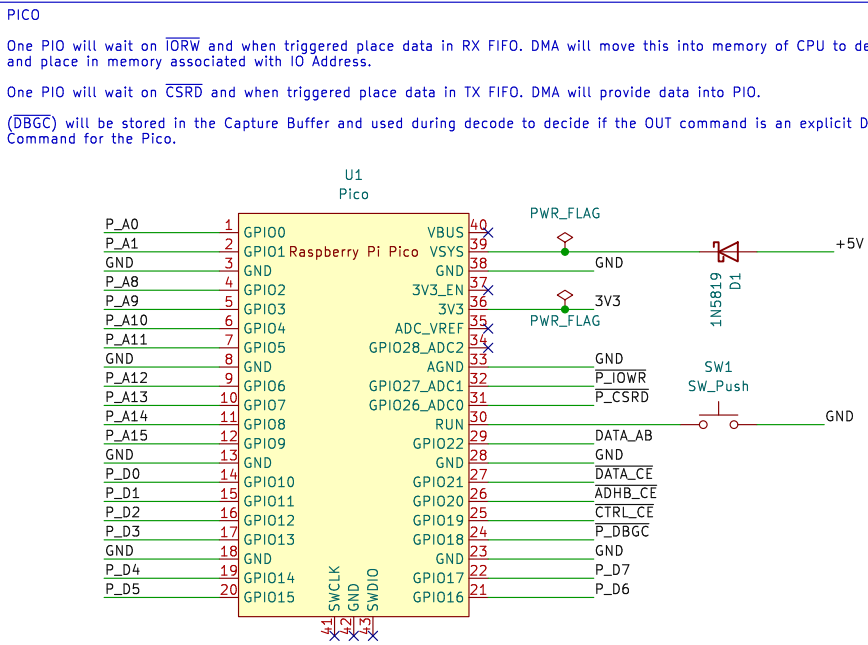
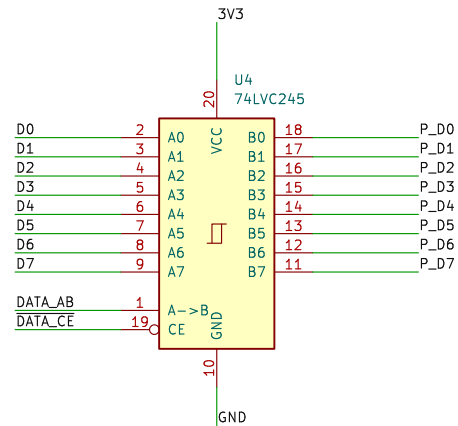


Note Direction Fixed as A to B

This transceiver buffers the high byte of the Data lines to and from the CPC.

The direction of data will be controlled by the Pico using Data Chip Enable (DATA_AB). By default this will be A to B and only on a IO Port Read will the Pico change this to B to A.

Data Chip Enable (DATA_CE) will be pulled low by the Pico when powered.



V2.0 Change GPIO to add DBGC for explicit selection of Debug Commands

Sheet: /
File: Amstrad-CPC-IO-Logger.kicad_sch

Title: RABS Amstrad CPC IO Data Logger

Size: A2	Date: 2026-02-15
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Rev: V2.0